Hibi

(5 pts) Brief paragraph describing your system and its requirements

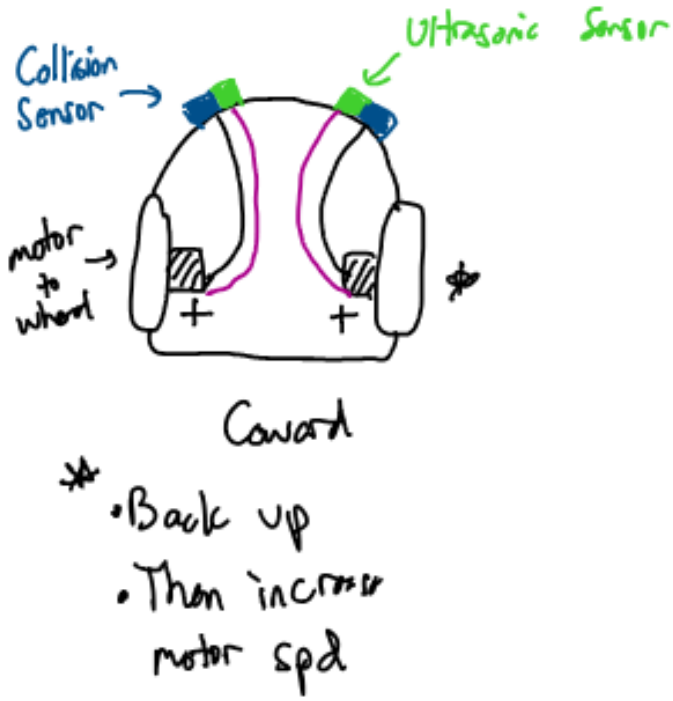
Inspired by Chapter 5 of Understanding Intelligence by Rolf Pfeifer, I built Hibi, an autonomous mobile robot designed for obstacle avoidance using proximity and collision sensors. Hibi integrates ultrasonic distance sensors and mechanical collision switches to perceive its environment and control two DC motors through an embedded microcontroller system. Its behavior is governed by a finite state machine that enables forward motion and avoidance maneuvers, while a neural-network–inspired control structure based on Hebbian learning strengthens associations between proximity sensing and avoidance actions. This allows the robot to gradually rely more on early proximity detection rather than collision events when navigating obstacles.

Requirements: (Refer to Datasheet and Resources Sections)

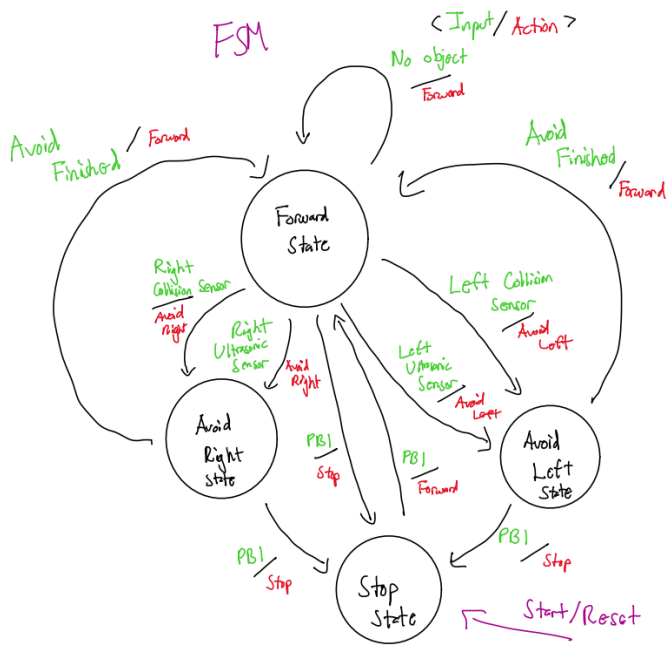
- Sensing subsystem
    - (HC-SR04) 2 Ultrasonic distance sensors for early obstacle detection
    - (SS-5GL2) 2 Mechanical limit switches for collision detection
- Control subsystem
    - (DFRduino Uno v3.0) 1 Arduino Uno microcontroller for sensor processing and decision making
    - DFRobot Prototype Shield for Uno
    - Embedded finite state machine and Hebbian-inspired control logic
- Actuation subsystem
    - 2 DC motors for robot locomotion
    - (TB6612FNG) Motor driver shield for controlling motor speed and direction
- Power subsystem
    - Battery supply for the Arduino and motor driver
- User interface
    - 1 Push Button for starting/stopping the robot
    - 1 Onboard LED for system status indication

(5 pts) Finite State Machine Diagram or table

Reference Diagram



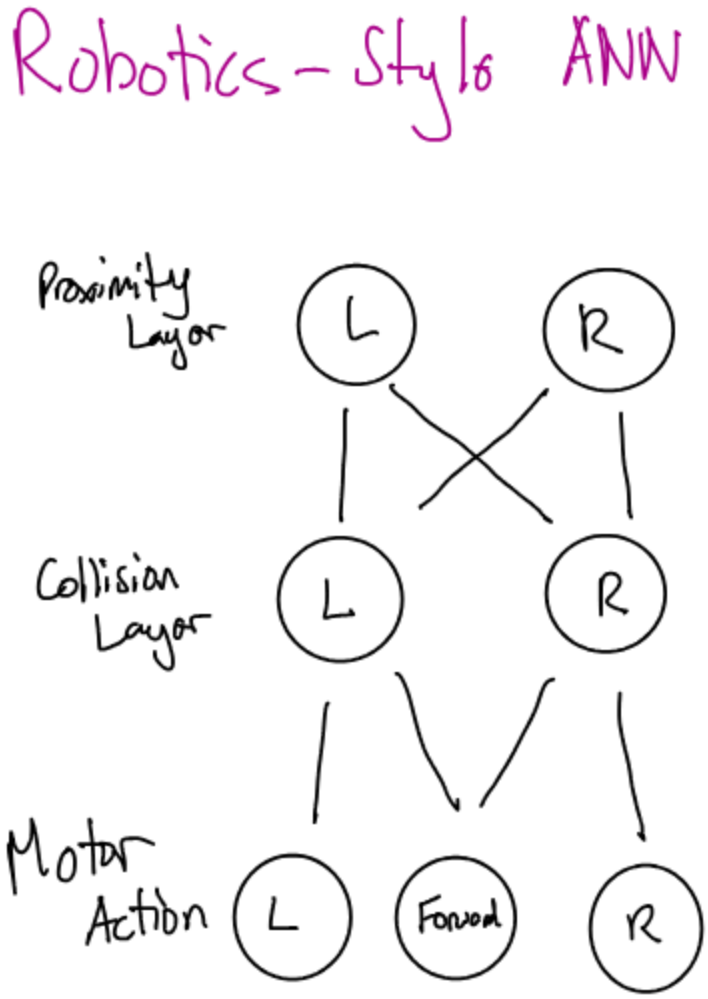
Finite State Machine Diagram



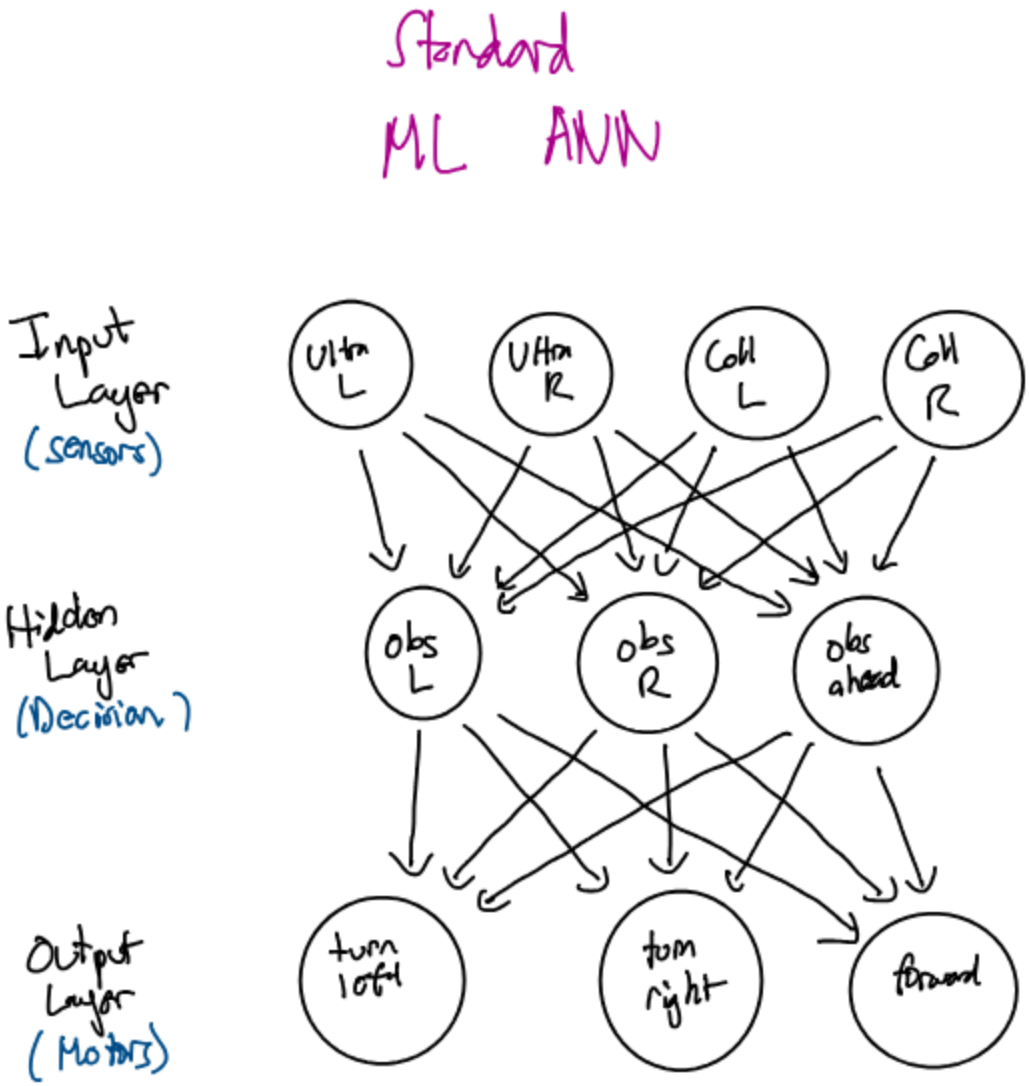
State Descriptions

- Forward State
  - ↳ Both motors forward
- Avoid Right State
  - ↳ Both motors reverse for 2 sec then turn left (Lmotor = idle, Rmotor = High)
  - ↳ If collision, ultrasonic weighting increases
- Avoid Left State
  - ↳ Both motors reverse for 2 sec then turn right (Rmotor = idle, Lmotor = High)
  - ↳ If collision, ultrasonic weighting increases
- Stop State
  - ↳ Turn off sensors + motors

Robotics Style ANN Diagram



Standard ML ANN Diagram



(5 pts) Documentation: An infographic summarizing your system and at least one photo and one video of your system in action. Cell phone images and video are fine, but production quality matters. Please try to use good lighting and steady camerawork.

Infographic:
https://drive.google.com/file/d/1Fx8aNa_9BHokyRSV4DMcMjhIMGPzt8rJ/view?usp=sharing

Video:
https://drive.google.com/file/d/11S7KAj6Iqm0KiBqjt94TlFJgFTxz4p3S/view?usp=sharing

Datasheets:

Limit Switch (SS-5GL2):
https://omronfs.omron.com/en_US/ecb/products/pdf/en-ss.pdf

- Actuator: Hinge Roller Lever
- Contact Form: SPDT

Ultrasonic Sonar Distance Sensor (HC-SR04):
https://handsontec.com/dataspecs/sensor/SR-04-Ultrasonic.pdf

Motor Driver Shield (TB6612FNG):
https://cdn-shop.adafruit.com/datasheets/TB6612FNG_datasheet_en_20121101.pdf

DFRobot Prototype Shield for Uno:
https://drive.google.com/file/d/1QQJZuXcxKEsXqXhCjPRjp93PmYUfy-Mz/view?usp=sharing

Resources:

Understanding intelligence Book by Rolf Pfeifer (Chap 5 pg. 139/160):
https://drive.google.com/file/d/1d1wppGQiDiBHhIOcZgl9_t_G0V4phKvB/view?usp=sharing

Arduino Uno Pinout:
https://docs.arduino.cc/resources/pinouts/A000066-full-pinout.pdf